Mini Paper — 1.3 Exchanging Data

Time: ~35 minutes · Total: 30 marks · Answer all questions.

Mark your work against the mark scheme at the end; aim for ~1 minute per mark.

Questions

Q1. Explain what is meant by *hashing* and give **two** reasons why hashing is used to store passwords rather than storing the passwords directly. **[3]** *(AO1)*

Q2. A company sends confidential data over the internet.

(a) State the difference between *symmetric* and *asymmetric* encryption. **[2]** *(AO1)*

(b) Give **one** advantage of using asymmetric encryption when two parties have never communicated before. **[1]** *(AO1)*

(c) Explain why the company hashes its stored user passwords but encrypts the data it transmits, referring to the key difference between hashing and encryption. **[1]** *(AO2)*

Q3. A row of pixels in a monochrome image is shown below, where each letter is a pixel colour:

`BBBBBBBWWWWBBBBBB`

(a) Show how *run length encoding (RLE)* would encode this row. **[2]** *(AO2)*

(b) State **one** type of data for which RLE would be a poor choice, and explain why. **[1]** *(AO2)*

Q4. A flat-file table storing patient appointments is being normalised.

`Appointment(AppointmentID, PatientID, PatientName, DoctorID, DoctorName, RoomNo)`

Here `DoctorName` and `RoomNo` are determined by `DoctorID`.

(a) State the rule a table must satisfy to be in *first normal form (1NF)*. **[1]** *(AO1)*

(b) The table is already in 2NF. Explain why it is **not** in *third normal form (3NF)*, and show the tables that result from normalising it to 3NF. **[3]** *(AO2)*

Q5. A `Member` table has the fields `MemberID`, `Surname`, `Town` and `JoinYear`.

Write an SQL statement to display the `MemberID` and `Surname` of all members whose `Town` is `'Leeds'` **and** who have a `JoinYear` before `2020`, sorted by `Surname` in descending order. **[4]** *(AO2)*

Q6. Describe the role of the **Transport** layer and the **Network/Internet** layer of the TCP/IP stack when data is sent. **[3]** *(AO1)*

Q7. Compare *packet switching* and *circuit switching*. **[3]** *(AO2)*

Q8. An online banking website lets customers log in and transfer money between accounts.

Discuss how *client-side processing* and *server-side processing* should be used for this website, and explain the role of **encryption** and **transaction processing (ACID)** in keeping it reliable and secure. Make a justified recommendation. **[6]** *(AO3)*